Entanglement in Levine's Hat Game: Finite and Infinite Strategies with One Bell Pair

github.com/emzeay/quantum-levine-hats

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1 Introduction and main results

In Levine's hat game [1], Alice and Bob each wear a stack of hats, independently colored black or white by fair coin flips. Alice sees Bob's hats but none of her own; Bob sees Alice's but none of his own. They agree on a strategy before the hats are drawn, then each selects a position in their own stack without communicating. They win if both selected hats are black. Each player individually has a $1/2$ chance of choosing black, but by coordinating their choices, they can make these successes occur together more often than the $1/4$ probability of two independent successes.

We give four-hat and five-hat strategies using one Bell pair. The four-hat strategy beats the proved classical optimum. The five-hat strategy exceeds the conjectured classical ceiling of $7/20$ for infinite stacks [2] and supports a single recurrence for every finite stack size. Its infinite extension uses the same Bell pair.

Theorem 1.1 (Four-hat strategy)**.** *There is a strategy with four hats per player that uses one Bell pair and succeeds with probability*

$$Q_4 = \frac{1517}{4352} > \frac{89}{256}.$$

The optimal classical success probability with four hats per player is $89/256$ [3], and Appendix A also gives a reproducible computational proof. Thus, the quantum advantage is $1/1088$.

Theorem 1.2 (Five-hat strategy)**.** *There is a strategy with five hats per player that uses one Bell pair and succeeds with probability*

$$Q_5 = \frac{1463}{4160} > \frac{7}{20}.$$

Classical five-hat strategies achieving $358/1024$ were known previously [3, 7]. The unrestricted SCIP computation in Appendix C reports the same optimum, allowing different strategies for Alice and Bob. This supplies computational evidence for a quantum advantage at five hats, but not yet a certified classical bound.

Theorem 1.3 (Finite extension)**.** *For every integer $h \geq 4$, write $h = 5k + r$ with $r \in \{1, \ldots, 5\}$ and $k \geq 0$. There is an explicit strategy with h hats per player that uses one Bell pair and succeeds with probability*

$$Q_h = L + (Q_r - L)256^{-k}, \qquad L = \frac{7781}{22100}. \tag{1}$$

Here $Q_1 = 1/4$, $Q_2 = 5/16$, $Q_3 = 11/32$, and Q_4, Q_5 are the values above. For every $h \geq 5$, the construction satisfies $Q_h \geq Q_5 > 7/20$.

For larger finite stacks and for infinite stacks, the comparison remains conditional on the $7/20$

conjecture [2]. The proved classical upper bound 81/224 [2, Section 3.1] is higher than the values of these larger-stack constructions.

Theorem 1.4 (Infinite extension)**.** *For countably infinite stacks, there is a measurable strategy using one Bell pair that inspects finitely many hats with probability* 1 *and succeeds with probability*

$$Q_\infty = \frac{7781}{22100} = \frac{7}{20} + \frac{23}{11050}. \tag{2}$$

The four-hat advantage comes from a biased form of the Clauser–Horne–Shimony–Holt (CHSH) inequality [8, 9]. The five-hat response table uses a larger collection of Bell correlations. The extension adapts the classical monochromatic-block recursion [2], with each player choosing a block before measuring their qubit at most once.

2 Game model and notation

Classical and quantum strategies use the same randomly assigned hats and winning condition. For an integer $h \geq 1$, let $x, y \in \{0,1\}^h$ describe Alice's and Bob's independent uniform stacks of h hats each. A value of 1 denotes black, and a value of 0 denotes white. Positions are numbered from left to right. Alice sees y and chooses $a \in \{1, \ldots, h\}$. Bob sees x and chooses b in the same range. The winning indicator is

$$W(x, y, a, b) = x_a y_b.$$

A deterministic classical strategy has $a = f(y)$ and $b = g(x)$. A randomized strategy uses shared or private randomness, independent of the hats, to choose a deterministic response pair. The success probability of a randomized strategy is the average over those deterministic strategies, so it cannot exceed the best deterministic value.

A quantum strategy uses a shared density operator ρ on $\mathcal{H}_A \otimes \mathcal{H}_B$, independent of the hats. Alice and Bob use local positive operator-valued measures (POVMs), $\{E_a^y\}_a$ and $\{F_b^x\}_b$, whose outcomes specify their selected hat positions. Each POVM consists of positive operators summing to the identity. The output probabilities are

$$P(a, b \mid y, x) = \operatorname{Tr}\big(\rho(E_a^y \otimes F_b^x)\big).$$

The success probability is the expectation of W over the hats and measurement outcomes.

By no-signaling, each player's distribution of selected positions is independent of their own hat string. Their selected hat is therefore black with probability 1/2, even when they share entanglement.

3 An explicit four-hat strategy

Alice and Bob share a Bell pair. Each chooses a binary measurement based on the other player's hats and uses its outcome to select a position in their own stack. Alice has three measurement settings and Bob has two. We give the measurements and response table, calculate the success probability, and show how biased CHSH produces the advantage.

3.1 Measurements and response table

Write the Pauli matrices and Bell state as

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad |\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

Alice measures one of A_0, A_1, A_2 and Bob measures one of B_0, B_1, defined by

$$A_0 = \frac{15Z - 8X}{17}, \quad A_1 = -Z, \quad A_2 = \frac{15Z + 8X}{17}, \qquad B_0 = Z, \quad B_1 = X. \tag{3}$$

These observables are Hermitian and square to the identity because $XZ = -ZX$ and $15^2 + 8^2 = 17^2$. Their outcomes are $+1$ and -1. For Alice, outcome s of setting A_i has projector $(I + sA_i)/2$, and similarly for Bob.

Table 1 completes the strategy. Alice uses the row for y and Bob uses the row for x. Each row gives a measurement setting, when one is needed, and the position to select for each outcome. For example, if Alice sees `0101` on Bob's head, she measures A_2 and selects her own second hat on outcome $+1$ or her own fourth hat on outcome -1. If both outcomes give the same position, then the player can select it without measuring.

Observed hats	Alice			Bob		
	Setting	$+1$	-1	Setting	$+1$	-1
`0000`	—	1	1	—	1	1
`0001`	—	4	4	—	4	4
`0010`	—	3	3	—	3	3
`0011`	—	3	3	B_1	3	4
`0100`	A_1	1	2	B_0	2	1
`0101`	A_2	2	4	—	4	4
`0110`	A_1	1	3	B_0	2	1
`0111`	A_1	4	3	—	4	4
`1000`	A_1	2	1	B_0	1	2
`1001`	A_1	2	4	B_0	1	2
`1010`	A_0	1	3	—	3	3
`1011`	A_1	3	4	—	3	3
`1100`	—	1	1	B_1	2	1
`1101`	—	2	2	B_0	1	2
`1110`	—	1	1	B_0	2	1
`1111`	—	1	1	—	1	1

Table 1. *Four-hat response table. The $+1$ and -1 columns give the position to select in the player's own stack for each measurement outcome. Positions are numbered from 1 to 4, left to right. A dash means the same position is selected for both outcomes, so no measurement is needed.*

All measurement projectors and the Bell density operator have rational entries, so the success probability can be calculated exactly using fractions alone.

3.2 Success probability

The success probability for this response table depends on five correlations. To derive the formula, we keep the table's measurement labels and hat choices fixed while allowing the measurements A_i, B_j themselves to vary. Write $\alpha_i = \langle A_i \rangle$ and $\beta_j = \langle B_j \rangle$ for the local means, and $C_{ij} = \langle A_i \otimes B_j \rangle$ for the correlations, with expectations taken in the shared state.

For fixed settings i, j, let $S, T \in \{+1, -1\}$ be the measurement outcomes. The indicator of $S = s, T = t$ is $(1 + sS)(1 + tT)/4$. Taking its expectation gives

$$P(s, t \mid i, j) = \frac{1 + s\alpha_i + t\beta_j + stC_{ij}}{4}. \tag{4}$$

For each pair of hat strings, we add the probabilities of the measurement outcomes that select two black hats. We then average over all 256 equally likely pairs of hat strings. When these contributions are added, the coefficients of every local mean α_i, β_j sum to zero. The winning probability therefore depends only on the five correlations below.

Lemma 3.1. *The success probability q for Table 1 is*

$$1024q = 324 + 4C_{00} - 2C_{01} - 24C_{10} + 4C_{20} + 2C_{21}. \tag{5}$$

Proof. Let $a_+(y), a_-(y)$ and $b_+(x), b_-(x)$ be the positions listed for Alice and Bob. Define

$$\begin{aligned} c_A(x,y) &= x_{a_+(y)} + x_{a_-(y)}, & d_A(x,y) &= x_{a_+(y)} - x_{a_-(y)}, \\ c_B(x,y) &= y_{b_+(x)} + y_{b_-(x)}, & d_B(x,y) &= y_{b_+(x)} - y_{b_-(x)}. \end{aligned}$$

Here c_A is the sum of Alice's two candidate hat colors, and d_A is their difference. Thus $(c_A + sd_A)/2$ gives the color she selects on outcome s. The same holds for Bob.

Multiplying the selected colors gives the winning indicator $(c_A + sd_A)(c_B + td_B)/4$. Taking its expectation over the measurement outcomes gives

$$\frac{c_A c_B + \alpha_i d_A c_B + \beta_j c_A d_B + C_{ij} d_A d_B}{4},$$

where i, j are the settings selected by the observed stacks.

We sum this expression over all 256 input pairs and collect terms by measurement setting. The constant coefficient is $\sum_{x,y} c_A c_B$. The coefficients of $\alpha_i, \beta_j, C_{ij}$ are the corresponding sums of $d_A c_B, c_A d_B, d_A d_B$. Write K_{ij} for the sum of $d_A d_B$ over input pairs selecting settings i, j. The table gives

$$\sum_{x,y} c_A c_B = 324, \qquad \text{all local-mean coefficients} = 0, \qquad K = \begin{pmatrix} 4 & -2 \\ -24 & 0 \\ 4 & 2 \end{pmatrix}.$$

Dividing by $4 \cdot 256$ gives the claimed success-probability formula. □

In the Bell state $|\Phi^+\rangle$,

$$\langle Z \otimes Z \rangle = \langle X \otimes X \rangle = 1, \qquad \langle Z \otimes X \rangle = \langle X \otimes Z \rangle = 0.$$

For example, substituting $A_0 = (15Z - 8X)/17$ and $B_0 = Z$ gives

$$C_{00} = \frac{15}{17}\langle Z \otimes Z \rangle - \frac{8}{17}\langle X \otimes Z \rangle = \frac{15}{17}.$$

Applying the same identities to all settings in (3) gives

$$(C_{ij}) = \begin{pmatrix} 15/17 & -8/17 \\ -1 & 0 \\ 15/17 & 8/17 \end{pmatrix}.$$

Equation (5) therefore gives

$$Q_4 = \frac{324 + 24 + 152/17}{1024} = \frac{1517}{4352}. \tag{6}$$

This proves Theorem 1.1. The next subsection explains why these correlations beat the classical bound.

3.3 Source of the quantum advantage

The expression

$$T = 4C_{00} - 2C_{01} + 4C_{20} + 2C_{21}$$

is a biased CHSH expression. The four terms involve two settings per player, with three positive coefficients and one negative coefficient, as in CHSH. In the corresponding biased CHSH game, these unequal magnitudes arise from choosing Alice's settings with equal probability and Bob's settings with probabilities 2/3 and 1/3. After relabeling Alice's settings, $T/12$ is the biased CHSH expression studied by Lawson, Linden, and Popescu [9].

For deterministic local outcomes $a_0, a_2, b_0, b_1 \in \{+1, -1\}$,

$$T = a_0(4b_0 - 2b_1) + a_2(4b_0 + 2b_1) \leq |4b_0 - 2b_1| + |4b_0 + 2b_1| = 8.$$

Since $b_0, b_1 \in \{+1, -1\}$, the two absolute values are 2 and 6, in either order. Their sum is therefore always 8. Randomized classical strategies are averages of deterministic assignments, so they satisfy the same bound. Since $-24C_{10} \leq 24$, classical strategies using this table succeed with probability at most $(324 + 24 + 8)/1024 = 89/256$. The fixed outcomes $a_0 = a_2 = b_0 = b_1 = +1$, $a_1 = -1$ give success probability 89/256, so this is the classical optimum within the response table.

The measurements in (3) attain $T = 152/17 > 8$ and $-24C_{10} = 24$, increasing the CHSH contribution while retaining the maximum contribution from the remaining term. The resulting success probability 1517/4352 exceeds 89/256. Since 89/256 is also the known optimum over all classical strategies [3], this establishes a quantum advantage for the four-hat game.

4 An explicit five-hat strategy

The five-hat construction follows the same pattern. Alice and Bob share $|\Phi^+\rangle$, choose a binary measurement from the observed stack, and turn its outcome into a hat position. Each needs seven active settings. The observables are

$$\begin{aligned}
A_0 = B_0 &= X, & A_1 = B_1 &= Z, \\
A_2 &= \frac{3Z + 4X}{5}, & B_2 &= \frac{3Z - 4X}{5}, \\
A_3 = B_3 &= \frac{4Z - 3X}{5}, & A_4 = B_4 &= \frac{4Z + 3X}{5}, \\
A_5 = B_5 &= \frac{5Z - 12X}{13}, & A_6 = B_6 &= \frac{5Z + 12X}{13}.
\end{aligned} \tag{7}$$

Each squares to the identity. As before, outcome s of observable A_i has projector $(I + sA_i)/2$. Here and below, an observed word is the binary string of hats visible to a player. Table 2 specifies the measurement and hat positions for every observed word.

To calculate the success probability, write $A_i = z_i Z + x_i X$ and $B_j = w_j Z + y_j X$. The Bell-state identities from Section 3 give $C_{ij} = z_i w_j + x_i y_j$. The local means vanish because the state has maximally mixed marginals and these observables are traceless.

Observed hats	Alice			Bob			Observed hats	Alice			Bob		
	Setting	$+1$	-1	Setting	$+1$	-1		Setting	$+1$	-1	Setting	$+1$	-1
00000	—	1	1	—	1	1	10000	—	5	5	—	5	5
00001	—	1	1	—	1	1	10001	A_2	5	1	B_4	1	5
00010	—	4	4	—	4	4	10010	A_3	5	4	B_0	4	5
00011	A_0	4	1	B_2	1	4	10011	A_0	4	1	B_0	4	5
00100	—	3	3	—	3	3	10100	A_1	5	3	B_0	3	5
00101	—	3	3	—	3	3	10101	A_1	5	3	B_6	3	5
00110	—	3	3	B_0	4	3	10110	A_1	5	3	B_0	4	5
00111	—	3	3	B_5	3	4	10111	A_1	5	3	B_0	4	5
01000	—	2	2	—	2	2	11000	—	2	2	—	2	2
01001	A_0	2	1	B_1	1	2	11001	A_6	2	1	B_1	1	2
01010	A_0	4	2	—	2	2	11010	A_5	2	4	—	2	2
01011	A_0	4	1	B_1	1	2	11011	A_0	4	1	B_1	1	2
01100	A_1	3	2	B_1	2	3	11100	—	2	2	B_3	2	3
01101	A_3	3	2	—	3	3	11101	A_0	2	5	B_0	3	5
01110	A_4	3	2	B_4	2	3	11110	A_0	4	2	B_1	2	4
01111	A_1	3	1	B_0	1	3	11111	—	1	1	—	1	1

Table 2. *Five-hat response table. The outcome columns give positions in the player's own stack, numbered from 1 to 5, left to right. A dash means the same position is selected for both outcomes, so no measurement is needed.*

Using the candidate-color sums and differences from Lemma 3.1, now summed over all 1024 five-hat deals, gives

$$4096q = 1319 + \sum_{i,j=0}^{6} K_{ij}C_{ij}, \qquad K = \begin{pmatrix} 29 & 1 & -4 & -3 & 6 & -6 & 5 \\ -1 & 28 & 2 & 4 & 7 & 1 & 1 \\ 4 & 2 & 1 & 0 & 0 & 0 & 0 \\ -6 & 7 & 0 & 0 & 1 & 1 & 0 \\ 3 & 4 & 0 & 0 & 0 & 0 & 1 \\ -5 & 1 & 0 & 1 & 0 & 1 & 0 \\ 6 & 1 & 0 & 0 & 1 & 0 & 1 \end{pmatrix}. \tag{8}$$

Here the constant 1319 is $\sum_{x,y} c_A c_B$, and each K_{ij} is the sum of $d_A d_B$ over deals selecting settings i, j. Equation (8) holds for any traceless qubit observables assigned to these labels on $|\Phi^+\rangle$.

Substituting the five-hat observables gives $\sum_{i,j} K_{ij}C_{ij} = 7897/65$, hence

$$Q_5 = \frac{1319 + 7897/65}{4096} = \frac{1463}{4160}.$$

This proves Theorem 1.2. We next use the four- and five-hat constructions as starting points for one recurrence.

5 Extending the strategy using one Bell pair

We adapt the classical construction of skipping monochromatic blocks described by Buhler et al. [2, Section 2.1]. Kallus [5] and Bouquet et al. [7] give related finite-to-infinite transformations. The following composition rule applies whenever the two strategies use the same shared state.

Lemma 5.1 (Block composition)**.** *Suppose strategies on m and t hats per player have success probabilities p and r. Assume both use the same shared state, with each player choosing at most one local measurement from their observed hats. There is then a strategy on $m+t$ hats using that state with success probability*

$$p - 4^{-m} + 4^{1-m}r. \tag{9}$$

Proof. Each player inspects the first m hats in the other player's stack. If they are nonmonochromatic, the player uses the m-hat strategy and selects a position in their own first block. Otherwise, they use the t-hat strategy on the tails. The players may choose different blocks. Both make their measurement only after deciding which rule to use.

Write the stacks as (u, x') and (v, y'), with first blocks $u, v \in \{0,1\}^m$. Compare the composed strategy with applying the m-hat rule to the first block alone. If both blocks are nonmonochromatic, its contribution is unchanged.

Suppose u is nonmonochromatic and v is monochromatic. Alice moves to the tail, while Bob selects in v. When v is all white, they lose. When v is all black, Bob succeeds automatically. No-signaling makes Alice's selected tail position independent of her own uniformly random tail, so she succeeds with probability $1/2$. This mixed case contributes $(2^m-2)/(2\cdot 4^m)$. The original first-block rule gives the same contribution, since each position is black in exactly $2^{m-1}-1$ nonmonochromatic words. The reversed mixed case is identical. This uses marginal probabilities, without assuming that the measurement outcomes are independent.

If both first blocks are monochromatic, the original rule wins only on the all-black pair, contributing 4^{-m}. The composed rule uses the tail strategy in all four monochromatic pairs, contributing $4^{1-m}r$. Replacing that contribution gives the stated success probability. □

For the finite family, the first block has five hats and uses Table 2. A player who sees a monochromatic block applies the same rule to the remaining hats until at most five remain. At that point, use the corresponding base strategy. The four- and five-hat bases are the constructions already given. The smaller bases are classical and ignore the Bell pair. For one hat, select it. For two hats, both players select positions $(1,2,1,1)$ on the lexicographically ordered observations $00, 01, 10, 11$. For three hats, use positions $(1,1,2,1,3,3,2,1)$ on observations $000, \ldots, 111$. Direct counting gives their scores $Q_1 = 1/4$, $Q_2 = 5/16$ and $Q_3 = 11/32$.

All bases therefore use the same state $|\Phi^+\rangle$, and skipping blocks requires no measurement. Each player measures their qubit at most once in the complete strategy. Applying Lemma 5.1 with $m = 5$ gives

$$Q_{h+5} = Q_5 - \frac{1}{1024} + \frac{Q_h}{256}, \qquad h \geq 1. \tag{10}$$

Its fixed point is

$$L = \frac{Q_5 - 1/1024}{1 - 1/256} = \frac{7781}{22100}.$$

Subtracting the fixed point L from the recurrence gives

$$Q_{h+5} - L = \frac{Q_h - L}{256}.$$

Iterating this identity within each residue class gives the closed formula for Q_h stated in Theorem 1.3. Each residue-class subsequence increases to L, since every base value $Q_1, \ldots, Q_5$ is smaller than L.

In particular, $Q_6 = Q_5$ and

$$Q_7 = Q_5 + \frac{1}{4096}, \qquad Q_8 = Q_5 + \frac{3}{8192}.$$

Every base has score at least $1/4$, and the recurrence preserves that lower bound. Equation (10) therefore gives $Q_h \geq Q_5 > 7/20$ for all $h \geq 5$. This proves Theorem 1.3. The one-Bell-pair construction now covers every finite size from four hats onward. Beyond the fixed base strategies, no additional response tables are needed.

6 Infinite stacks

The finite family converges to $L = 7781/22100$. We now show that the same rule on infinite stacks attains this value. Each player finds the first nonmonochromatic five-hat block in the other player's stack and applies Table 2. If that is block r, they select the indicated position within block r of their own stack.

Model the stacks as $x, y \in \{0,1\}^{\mathbb{N}}$ with all colors independent and fair. Two of the 32 possible five-hat blocks are monochromatic, so a block is skipped with probability $1/16$. Let T be the index of the first nonmonochromatic observed block, counted from one. Independence gives

$$\mathbb{P}(T > k) = 16^{-k}, \qquad k \geq 0. \tag{11}$$

Thus each player inspects finitely many hats with probability 1. On the event of probability 0 that no such block exists, assign setting zero and position one.

For each finite r, the event $T = r$, the chosen measurement, and the rule assigning outcomes to positions depend only on the first $5r$ observed hats. The rules are measurable because they are specified on countably many events determined by finite prefixes. The fallback event $T = \infty$ is measurable too, as the intersection of the events $T > k$.

Let $F_\infty(x, y)$ be the winning probability for fixed stacks, averaged over measurement outcomes, and let $F_k(x, y)$ be the winning probability of the finite $5k$-hat construction. The final five-hat block uses the base strategy even if it is monochromatic. Both are measurable and take values in $[0, 1]$, and $\mathbb{E}F_k = Q_{5k}$. If both players find a nonmonochromatic block within these k blocks, the finite and infinite strategies use identical measurements and positions. Consequently,

$$|\mathbb{E}F_k - \mathbb{E}F_\infty| \leq \mathbb{P}(T_A > k \text{ or } T_B > k) \leq 2\,16^{-k}.$$

Taking $k \to \infty$ and substituting the closed formula for Q_{5k} proves Theorem 1.4.

The expected inspection cost is finite as well. With k full blocks available,

$$\mathbb{E}\min(T, k) = \sum_{j=0}^{k-1} 16^{-j} = \frac{16}{15}(1 - 16^{-k}).$$

For infinite stacks this gives $\mathbb{E}T = 16/15$ blocks. Reading each inspected block in full therefore requires an average of $16/3$ hats per player.

7 Related work and scope

The four-hat construction combines the biased CHSH analysis of Lawson, Linden and Popescu [9] with an explicit response table for Levine's hat game. The five-hat table uses a larger correlation expression. Both constructions fit the classical block recursion described by Buhler et al. [2], and the same Bell pair supports the finite and infinite extensions.

Buhler et al. [2, Section 2.1] attribute the discovery of the classical $7/20$ block strategy in 2014 to Larry Carter, Jay-C Reyes, Joel Rosenberg and Mark Tiefenbruck, and to Jonathan Kariv and

Dmytro Yeroshkin. Kallus [5] gives a general finite-to-infinite block transformation in Aubrun's MathOverflow discussion [4]. Recent work on classical strategies for Levine's game includes the Fourier bounds of Heilman and Tamuz [6] and the recursive strategies of Bouquet et al. [7].

The four-hat classical optimum of 89/256 was established earlier. Wagon [3] reports a computational proof with Roche that includes asymmetric strategies, and attributes a separate hand proof to Stromquist. Aubrun's MathOverflow discussion [4] also records finite-stack strategies and Carlsson's independent CPLEX calculation. The original computational and hand proofs reported by Wagon have not been located. Appendix A supplies a reproducible computational proof of the unrestricted four-hat bound.

Wagon [3] reports 358/1024 as optimal among symmetric five-hat strategies, with independent confirmation by Rob Pratt. Bouquet et al. [7] give five-hat constructions with that score and a five-hat recursion attaining 7/20 on infinite stacks. The present Bell-pair strategy improves those achievable probabilities. The unrestricted classical computation in Appendix C allows different strategies for Alice and Bob.

Related quantum work includes the stabilizer-code construction of Zhao et al. [10]. Their public example and references indicate a version of Ebert's color-guessing game [11], in which each participant wears one hat and guesses its color or passes. The players win if at least one guess is correct and none are wrong. Lenstra and Seroussi [12] study this game's connection to covering codes. Levine's game instead asks both players to select a black hat from their own stack, so the two constructions address different winning conditions. The full article by Zhao et al. has not been obtained, so the comparison remains incomplete.

The unrestricted quantum optima remain undetermined for four and five hats. The four-hat separation uses a proved classical bound, while the five-hat comparison additionally depends on the computational upper bound in Appendix C. For larger finite stacks and infinite stacks, the strategies exceed 7/20 but do not establish unconditional quantum advantages.

A The unrestricted classical four-hat optimum

Wagon [3] reports the classical optimum 89/256, including asymmetric strategies. The computation below verifies this bound by enumerating Alice's strategies and finding Bob's best response to each.

Proposition A.1. *With four hats per player, every classical strategy succeeds with probability at most* 89/256*, and this value is attained.*

Proof. It suffices to consider deterministic strategies $f, g\colon \{0,1\}^4 \to \{1,2,3,4\}$. Let $N = \{0,1\}^4 \setminus \{0000, 1111\}$ be the 14 nonmonochromatic strings. First consider the input pairs with at least one monochromatic stack. An all-white stack makes winning impossible. If Alice's stack is all black and Bob's stack lies in N, Bob selects a fixed position $g(1111)$, which is black in exactly seven strings of N. These pairs therefore contribute seven wins, regardless of the strategies. Reversing the players contributes seven more, and the pair of all-black stacks contributes one. The total contribution from monochromatic stacks is thus always 15 wins.

It remains to count wins when both stacks lie in N. Fix Alice's choices $f(y)$ for all $y \in N$. For a given $x \in N$, Bob's choice b wins on exactly $\sum_{y\in N} x_{f(y)}y_b$ of these inputs. He can maximize this count separately for each x, so his best response gives the total number of wins

$$M(f) = 15 + \sum_{x\in N} \max_{b\in\{1,2,3,4\}} \sum_{y\in N} x_{f(y)}y_b.$$

Alice's choices on monochromatic inputs do not change the fixed contribution of 15. Maximizing $M(f)$ over the 4^{14} maps $f\colon N \to \{1,2,3,4\}$ therefore covers all pairs of deterministic strategies, including asymmetric ones.

To enumerate the maps, order N lexicographically as $y_0, \ldots, y_{13}$. For each integer $0 \le c < 4^{14}$, write $c = \sum_{j=0}^{13} d_j 4^j$, with $d_j \in \{0,1,2,3\}$, and set $f_c(y_j) = d_j + 1$. Evaluating the displayed formula for all $4^{14} = 268{,}435{,}456$ maps using integer arithmetic gives $\max_c M(f_c) = 89$. This gives an upper bound of $89/256$ on the success probability. The fixed outcomes $a_0 = a_2 = b_0 = b_1 = +1$, $a_1 = -1$ in Table 1 attain this value, as calculated in Section 3.3. Randomized strategies average deterministic success probabilities and obey the same bound. □

B An optimal refinement of the four-hat measurements

Keeping the response table fixed, we now optimize its measurements by specializing the biased-CHSH bound of Lawson, Linden and Popescu [9]. The input-to-setting and outcome-to-position maps remain those of Table 1, so inputs with the same setting label must use the same measurement.

Proposition B.1. *Fix the input-to-setting assignments and outcome-to-position maps of Table 1. Allow arbitrary finite-dimensional shared states, independent of the hats, and arbitrary binary local measurements, with one measurement assigned to each setting label. The largest success probability in this class is*

$$q_{\text{table}} = \frac{87 + \sqrt{5}}{256}.$$

Proof. Represent the shared state by a purification $|\psi\rangle$ and the binary measurements by Hermitian contractions A_i, B_j. This includes binary POVMs, since the difference of their two effects has operator norm at most one. Recall the biased CHSH expression $T = 4C_{00} - 2C_{01} + 4C_{20} + 2C_{21}$ from Section 3.3. Grouping its terms by Alice's setting and applying Cauchy–Schwarz gives

$$\begin{aligned} T &\le \|(I \otimes (4B_0 - 2B_1))|\psi\rangle\| + \|(I \otimes (4B_0 + 2B_1))|\psi\rangle\| \\ &\le \sqrt{2\langle\psi|I \otimes (32B_0^2 + 8B_1^2)|\psi\rangle} \le 4\sqrt{5}. \end{aligned}$$

The second inequality uses $(u+v)^2 \le 2(u^2+v^2)$ for the two norms. When their squares are added, the mixed products of B_0 and B_1 cancel. The last inequality uses $B_j^2 \le I$.

The remaining correlation term satisfies $-24C_{10} \le 24$. Substituting both bounds into the success-probability formula gives

$$q \le \frac{324 + 24 + 4\sqrt{5}}{1024} = \frac{87 + \sqrt{5}}{256}.$$

Both bounds are attained simultaneously on $|\Phi^+\rangle$ by

$$A_0 = \frac{2Z - X}{\sqrt{5}}, \quad A_1 = -Z, \quad A_2 = \frac{2Z + X}{\sqrt{5}}, \qquad B_0 = Z, \quad B_1 = X. \qquad \square$$

This maximum applies only to the fixed response table. Strategies that change the table or let a measurement depend further on the observed hats are outside this class.

C The classical five-hat computation

This appendix records the mixed-integer computation returning 358/1024 for unrestricted classical strategies. It also gives an attaining strategy that can be checked directly. The upper bound is the optimizer's conclusion; an independent exact certificate is not yet included.

Let $N_5 = \{0,1\}^5 \setminus \{00000, 11111\}$. The monochromatic cases contribute $15 + 15 + 1 = 31$ wins, by the same counting argument as Appendix A. For each $y \in N_5$ and $a \in \{1, \dots, 5\}$, introduce a binary variable $F_{y,a}$ indicating Alice's chosen position. Define $G_{x,b}$ for Bob in the same way and impose

$$\sum_{a=1}^{5} F_{y,a} = 1, \qquad \sum_{b=1}^{5} G_{x,b} = 1.$$

For each $x, y \in N_5$, set

$$U_{xy} = \sum_a x_a F_{y,a}, \qquad V_{xy} = \sum_b y_b G_{x,b}.$$

These are binary because exactly one position variable equals 1 for each observation. A continuous variable $W_{xy} \in [0,1]$ is forced to equal $U_{xy}V_{xy}$ by

$$W_{xy} \le U_{xy}, \qquad W_{xy} \le V_{xy}, \qquad W_{xy} \ge U_{xy} + V_{xy} - 1.$$

Maximizing $31 + \sum_{x,y \in N_5} W_{xy}$ therefore gives the classical winning count. The formulation has 300 binary position variables and 900 continuous winning variables and does not require Alice and Bob to use the same function.

For symmetry reduction, order N_5 lexicographically and impose

$$F_{y,a} \le \sum_{z<y} F_{z,a-1}, \qquad a = 2, \dots, 5.$$

These constraints require output positions to appear in first-occurrence order. Every strategy pair has an equivalent pair of this form. Permute Alice's physical hat positions so that her first distinct output is 1, the next is 2, and so on, then apply the inverse permutation to Bob's observed input. This relabeling permutes the equally likely deals and preserves both selected hat colors.

The model also uses the valid ceiling of 369 wins. If a five-hat classical strategy wins C deals, its monochromatic-block extension has infinite success probability $(C-1)/1020$ by Lemma 5.1. The published classical upper bound 81/224 [2, Section 3.1] therefore gives

$$C \le \left\lfloor 1 + 1020 \frac{81}{224} \right\rfloor = 369.$$

SCIP 10.0.2 [13], with feasibility tolerance 10^{-8} and one solving thread, returned matching primal and dual bounds of 358 after 445,277 nodes. The model's feasible strategies and this upper-bound computation are separate from the exact check of the following witness.

Read the successive digits below as positions chosen on observations $00001, \dots, 11110$, in lexicographic order. On 00000 and 11111, select position 1.

Alice `12131314141414122222224541454`,

Bob `12233221111111552255225541554`.

Evaluating these functions on all 1024 deals gives exactly 358 wins. The upper bound still relies on the completed optimizer run. The quantum five-hat advantage follows if that reported upper bound is accepted, and the weaker ceiling of 360 would already suffice.

D An irrational refinement of the five-hat measurements

The rational five-hat observables are convenient for exact arithmetic, but Alice can improve the score with Bob's observables held fixed. Keep the state $|\Phi^+\rangle$, Table 2, and Bob's five-hat measurements. For each i, define

$$V_i = \sum_{j=0}^{6} K_{ij}(w_j, y_j) = \frac{1}{65}(r_i, s_i),$$

where (w_j, y_j) are Bob's Z, X coefficients. The integer pairs are

i	0	1	2	3	4	5	6
r_i	40	2520	169	532	285	142	142
s_i	3104	−52	208	−411	255	−424	489

.

Replace Alice's observable A_i by

$$\widehat{A}_i = \frac{r_i Z + s_i X}{\sqrt{r_i^2 + s_i^2}}. \tag{12}$$

These are valid binary projective measurements, since each is a unit combination of Z and X. They give the exact success probability

$$q_{\text{alg}} = \frac{1}{4096}\left(1319 + \frac{1}{65}\sum_{i=0}^{6}\sqrt{r_i^2 + s_i^2}\right) > \frac{1463}{4160}. \tag{13}$$

Indeed, for any unit vector u_i in the Z, X plane specifying Alice's observable, its contribution to the correlation sum is $u_i \cdot V_i \le \|V_i\|$. The refined observables attain equality for every setting. The improvement is strict because the original vector $(3/5, 4/5)$ for A_2 is not parallel to $V_2 = (13/5, 16/5)$. Its contribution rises from $103/25$ to $\sqrt{17}$, and none of the other contributions decreases.

This optimizes Alice's observables within the Z, X plane, with Bob's measurements and the Bell state fixed. The joint optimum over both players' measurements remains undetermined for this five-hat table. Substituting q_{alg} for Q_5 in the block composition also gives an infinite-stack strategy with success probability

$$\frac{1024 q_{\text{alg}} - 1}{1020} > \frac{7781}{22100},$$

using the same Bell pair.

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